

Healthcare amenities, housing prices and senior migration

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1 Reminder of the research areas

Two broad classes of ideas:

1. How are healthcare amenities capitalized into housing prices?
2. How does senior migration drive up housing prices? If so, is healthcare quality a primary force attracting them? Are cities doing anything to try to attract seniors?

2 Suggestions from last meeting / Summary of what I've done since last time

1. Read:

- Read Alexander and Richards (2023), on the effects of rural hospital closures. Done.
- Check out Mathilde Muñoz recent work to get an idea of what has been done already: (Kalin et al., 2024) on international migration, and (Badilla Maroto et al., 2024), on domestic migration in France. Done, see discussion below.
- Two papers on how much people value healthcare: (Finkelstein et al., 2019b) and (Finkelstein et al., 2019a) - Still need to do this.

2. Institutional details and anecdotal data to collect:

- Is anyone saying they will retire to any of the Mayo Clinic locations (including Florida and Arizona locations)? - Yes, some evidence of this below.
- Is Mayo Clinic trying to build more housing for their people? - See below.
- How have house prices moved in Rochester since the 2013 announcement of DMC? - See below.

3. Statistical analyses:

- How did age composition (share 65+) change in places with (high HC vs low HC initial shares), which lost manufacturing? In other words, did having high initial health shares buffer the potential losses induced by manufacturing declines? See below.
- How are house prices being influenced by rural hospital closures (basically a slight extension of Alexander and Richards (2023))? See below.

3 Institutional details

3.1 How do people choose where to retire?

A survey by the insurance company, TransAmerica, listed the following top factors seniors reported considering when choosing where to retire: affordable cost of living (65%); proximity to family and friends (61%); access to quality health care (49%); low crime rates (48%); and good weather (42%). This is consistent with evidence from (Badilla Maroto et al., 2024), that pensioners tend to move to poorer areas. A stylized fact from that paper is that only the lower 2/3 of cantons, by GDP per capita, have net inflows of pensioners.

Also a fun fact is Mayo Clinic operates its own retirement [home](#).

Effect of Mayo Clinic build out on house prices (in Rochester)

We discussed trying to see if large build-outs, like the Destination Medical Center in Rochester, MN (which began in 2013), might have had an effect on house prices. I haven't run the DiD, but Rochester does not appear to show much of a differential gain relative to intuitive control groups. For such a gradual roll-out, identification seems tricky, as we discussed.

Is Mayo Clinic building out housing for its workers?

We discussed this question because the Destination Medical Center build-out has likely increased the number of jobs, and thus demand for housing, quite a lot in Rochester.

The answer: Not directly, though they are giving money to local groups, some of which support housing. This seemed to me more like surplus sharing to keep the local populations happy, rather than a direct investment in housing. For example, of the 14.25M dollars they are giving to Rochester, 10M is going to Rochester Public Schools. The quote below describes the \$22M they spent on communities they have locations in.

Mayo Clinic cares for its patients, communities and each other, and Mayo's Bold. Forward. strategy supports this care through work to Cure, Connect, and Transform healthcare for all. As this strategy successfully unfolds, this year Mayo Clinic is able to invest an additional \$22 million in community organizations addressing pressing needs related to housing, food insecurity, access to healthcare, and safe places for learning and youth enrichment. Mayo Clinic's \$22 million commitment will be distributed to organizations serving Rochester, Jacksonville, Phoenix and various Mayo Clinic Health System locations.

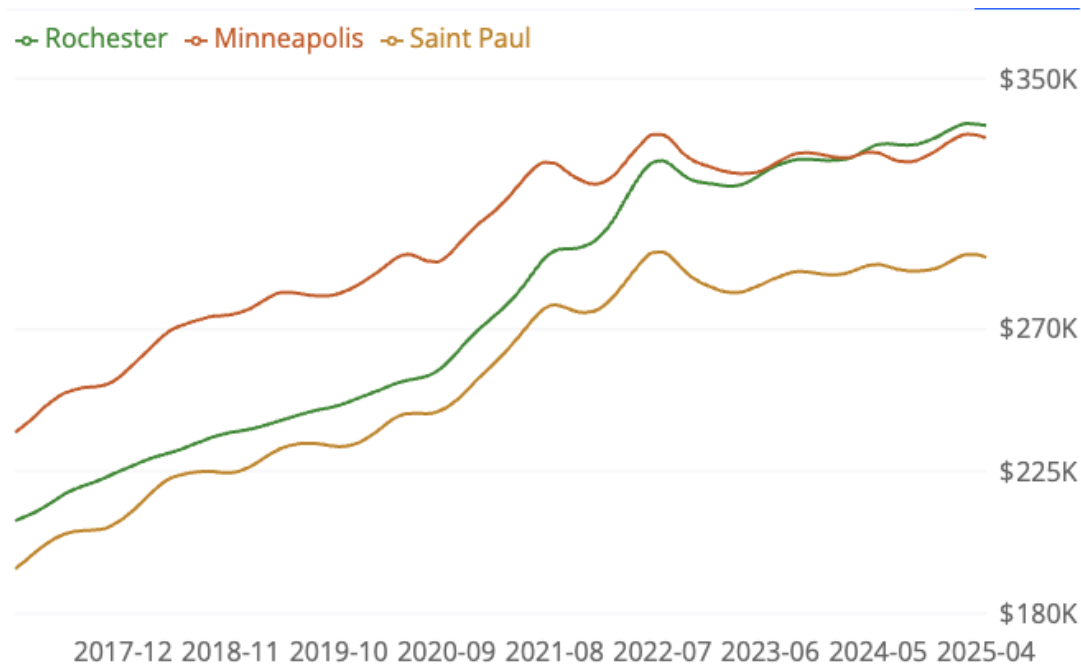


Figure 1: House prices in Rochester, MN, relative to Minneapolis, MN and St. Paul, MN, from 2017-2025. Source: Zillow Home Value Index.

4 How did age composition change in places with (high vs. low HC initial shares) that lost manufacturing?

As a reminder, you suggested I look at whether older folks are moving in at higher rates to high vs. low healthcare places, as an initial test. The specific way you suggested I operationalize this was to see if initial healthcare employment share mediated the effect of manufacturing decline at all.

Procedure:

1. Use employment data from county business patterns, and demographic data from census.
2. Classify healthcare as: NAICS 62 (Health Care and Social Assistance)
3. Classify manufacturing as: NAICS 31-33 (Manufacturing), calculate losses in employment shares from 2000 - 2010.
4. Bucket counties by their quartile of loss of manufacturing share, including those that gained, and plot the bivariate relationship in Figure 2.

It does not seem like high initial healthcare shares are positively correlated with changes in the over 65 share. In fact, there is a robust negative correlation, which makes me worry if this could be mechanical, i.e. the places with lots of healthcare already had a large 65+ share in 2000, and hence didn't have much room to go up. Long differences might not be the right way to think about

this, so I also look at cross sectional scatters in the appendix in Figure (5) and (6). These display much more variation in the dependent variable and have the expected positive correlation.

Looking across panels, the main effect of initial healthcare employment is clearly negative. But there is a significant interaction between manufacturing loss and healthcare share, as shown in the corresponding regression in Table 3.¹ Interpretation on this interaction:

- **How does the interaction modify the slope on manufacturing loss?** The more health intensive a county already was in 2000, the less negative the coefficient of losing manufacturing on 65+ share becomes. The sign flips from negative to positive once the initial share is above ≈ 0.068 , which is of course true for most counties.
- **How does the interaction modify the slope on health employment in 2000?** A county with no manufacturing loss sees a negative association between initial health employment share and subsequent senior-share growth. But if its share of manufacturing employment falls enough - 32 pp or more (so only a very small number of counties) - that association turns positive.

Which places **did** become older over this period? I detail these in the below table; surprisingly to me, Florida doesn't show up, and I can't tell a coherent story from these examples. For reference, the US as a whole went from 12.4% to 13.0% share of population 65+ over this period.

Table 1: Top 10 Counties by Increase in Elderly Population Share (2000-2010)

	County	Pop. (2000)	Pop. (2010)	Pct. 65+ (2000, %)	Δ Share 65+ (pp)
1	Beaufort County, South Carolina	120,937	162,233	15.51	4.85
2	Santa Fe County, New Mexico	129,292	144,170	10.75	4.37
3	Livingston County, Michigan	156,951	180,967	8.31	3.65
4	Marin County, California	247,289	252,409	13.52	3.20
5	Douglas County, Oregon	100,399	107,667	17.82	3.15
6	Jefferson County, Colorado	527,056	534,543	9.64	2.97
7	Douglas County, Colorado	175,766	285,465	4.17	2.96
8	Washington County, Minnesota	201,130	238,136	7.59	2.90
9	Mohave County, Arizona	155,032	200,186	20.47	2.84
10	Sussex County, New Jersey	144,166	149,265	9.12	2.84

Note: Limited to counties with population $\geq 100,000$ in 2000.

¹The main effect is visualized in Figure 4

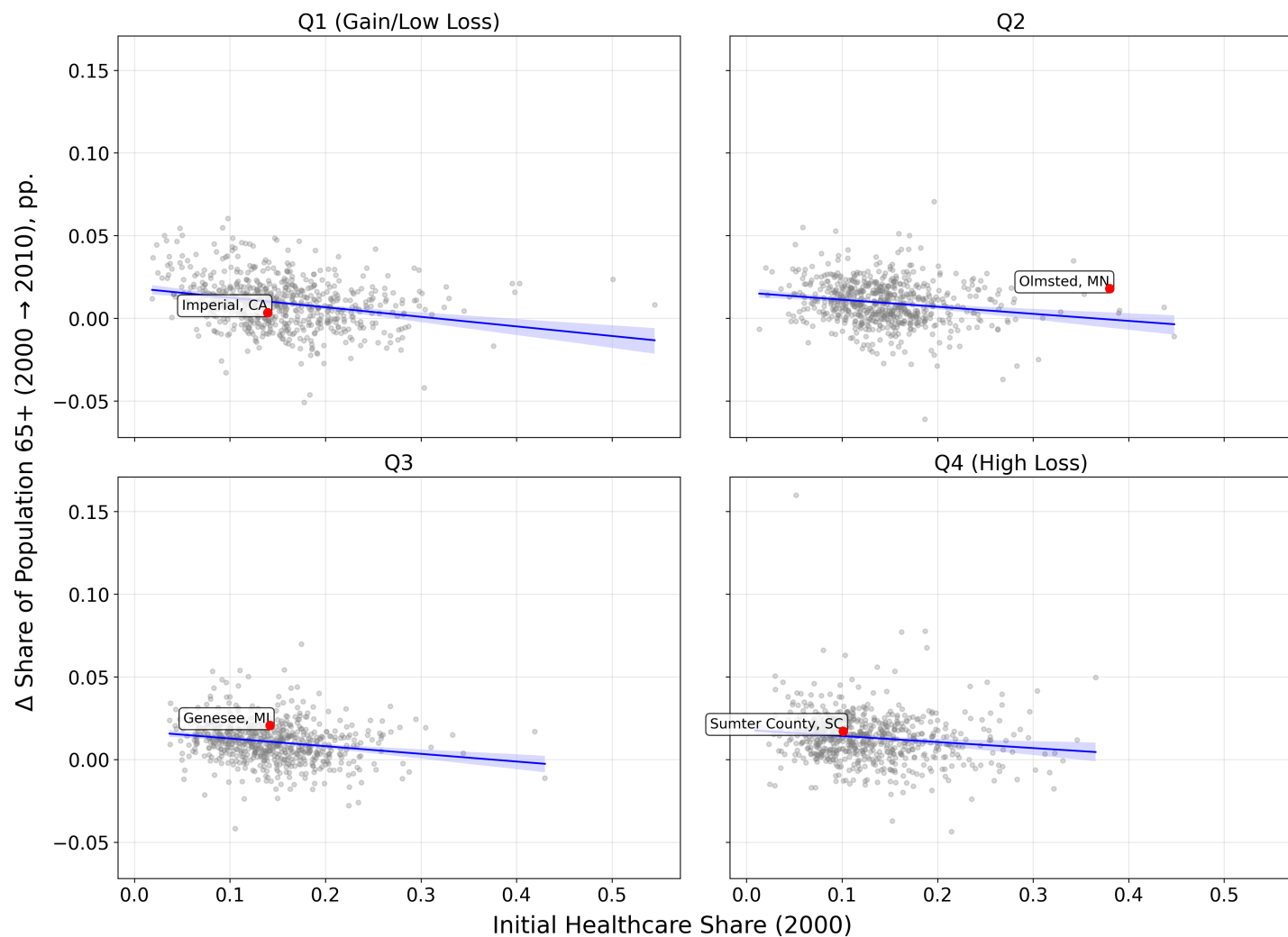


Figure 2: Panels respond to quartile of manufacturing employment share decline. Y-axis is change in share of population 65+ by county. X-axis is share of employment in healthcare in 2000. Restricts to counties with non-zero manufacturing and healthcare share in 2000. Changes are from 2000-2010. (CBP and Census)

5 Rural hospital closures and house prices

To get traction on the question of how healthcare amenities are capitalized into housing prices, I start with the extensive margin (going from 1 to 0), following the study of rural hospital closures in (Alexander and Richards, 2023).

I run a staggered DiD and treat county-quarters as observations.

Disclaimers and overview of data

1. At best, this would tell us the net effect of rural hospital closures on house prices, which includes both employment effects and healthcare amenities, so isn't a very clean test of the mechanism I am interested in.
2. I am using non-treated counties as a control. The control group used in (Alexander and Richards, 2023) were all rural counties with no hospital closures, which they licensed from the American Hospital Association.
3. I use the same rural hospital closure data as (Alexander and Richards, 2023), from the Sheps Center at UNC. Closure dates are summarized in Figure 8.
4. I am using the Zillow Home Value Index for all homes from January 2000 to April 2025, smoothed and seasonally adjusted. This is supposed to be the value of a typical home in the 35th-65th percentile for that county. This is the first time I am using this data, so I still need to check if this is a good measure (e.g. how they are doing the smoothing / seasonal adjustment).

Empirical Specification

1. **Unit of Observation:** Each observation in the dataset represents a county-quarter pair.
2. **Treatment Group:** Rural counties that experienced exactly one hospital closure during the sample period. (I drop three counties that experienced multiple)
3. **Control Group:** All other counties in the US.

$$ZHVI_{it} = \sum_{q=-4, q \neq -1}^4 \beta_q \cdot \mathbf{1}(t - t_i^* = q) + \alpha_i + \gamma_t + \epsilon_{it} \quad (1)$$

where:

- $ZHVI_{it}$ is the Zillow Home Value Index for county i in quarter t
- t_i^* is the quarter of hospital closure for treated county i
- $\mathbf{1}(t - t_i^* = q)$ is an indicator equal to 1 if quarter t is q quarters relative to the closure

- α_i are county fixed effects
- γ_t are quarter fixed effects

The presence of pre-trends, and lack of precision preclude much of an interpretation. Raw means are shown in appendix Figure 9

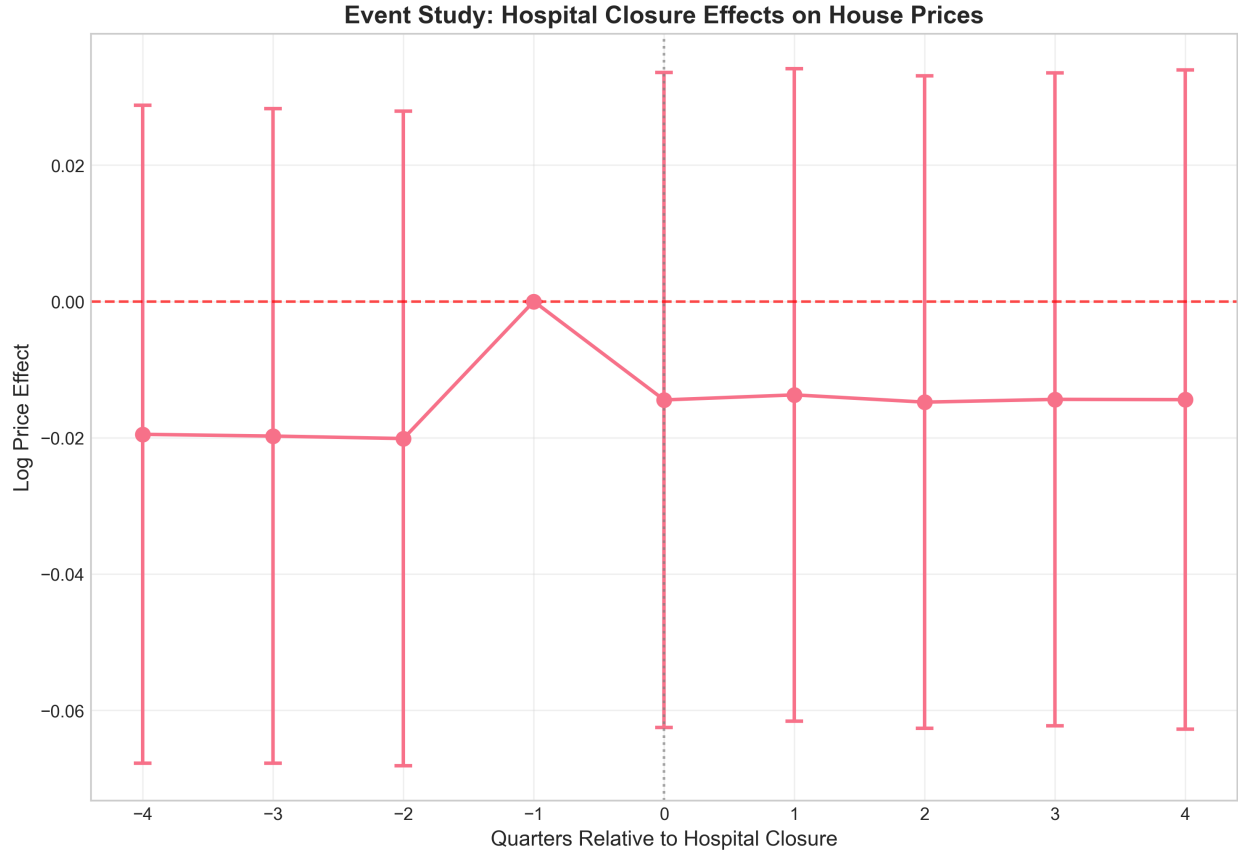


Figure 3: Event study coefficients for the effect of rural hospital closures on house prices, using a county-quarter panel. The omitted reference period is one quarter before the closure. The shaded areas represent 95% confidence intervals.

6 What research questions survived these analyses?

1. Does senior migration to high healthcare amenity places drive up rents? Status: Some puzzles here in the long-differences. The existing work seems to be very reduced form, so I am optimistic on continuing on this.
2. How do healthcare amenities influence rent? Status: data seems to exist to answer this question, it seems worth continuing, but the objection that multiple things are happening at once in the rural-hospital closure setting is well-taken.

A Appendix Figures and Tables

Table 2: Summary Statistics of Variables used in Manu. / HC. analysis

	Mean	Std. Dev.	Min	Q1	Median	Q3	Max	N
Δ Share of Pop. 65+ (2000-10)	0.011	0.014	-0.061	0.002	0.011	0.018	0.160	2628
Manu. Loss (Δ Share, 2000-10)	0.085	0.103	-0.323	0.023	0.059	0.113	0.658	2628
Δ HC Share (2000-2010)	0.021	0.069	-0.448	0.009	0.033	0.054	0.246	2628
HC Share (2000)	0.143	0.059	0.009	0.101	0.137	0.174	0.545	2628
Share of Pop. 65+ (2000)	0.144	0.039	0.023	0.119	0.142	0.166	0.347	2628
Total Pop. (2000, thousands)	104.631	314.327	1.902	16.349	32.845	75.792	9519.338	2628

Table 3: Regression of Change in Share of Population 65+ (2000-2010) on Covariates

	DV: Change in Share 65+, 2000-10
ManufLoss	-0.01013* (0.00600)
HealthChange	0.00380 (0.00389)
HealthShare_2000	-0.04718*** (0.00593)
ManufLoss:HealthShare_2000	0.14933*** (0.04438)
Share65_2000	-0.05598*** (0.00721)
TotalPop_2000_Thousands	-0.00001*** (0.00000)
Intercept	0.02515*** (0.00122)
R-squared	0.08286
R-squared Adj.	0.08076
N	2628

Standard errors in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 4: Top 10 Counties by Increase in Elderly Population Count (2000-2010)

	County	Pop. (2000)	Pop. (2010)	Pop. 65+ (2000)	Δ Count 65+
1	Los Angeles County, California	9,519,338	9,818,605	926,673	139,026
2	Maricopa County, Arizona	3,072,149	3,817,117	358,979	103,662
3	Harris County, Texas	3,400,578	4,092,459	252,895	80,592
4	Clark County, Nevada	1,375,765	1,951,269	146,899	73,546
5	Orange County, California	2,846,289	3,010,232	280,763	68,914
6	Riverside County, California	1,545,387	2,189,641	195,963	62,622
7	Tarrant County, Texas	1,446,219	1,809,034	120,585	40,800
8	San Diego County, California	2,813,833	3,095,313	313,750	37,675
9	Santa Clara County, California	1,682,585	1,781,642	160,527	36,417
10	San Bernardino County, California	1,709,434	2,035,210	146,459	34,889

Note: Limited to counties with population $\geq 100,000$ in 2000.

Table 5: Bottom 10 Counties by Change in Elderly Population Share (2000-2010)

	County	Pop. (2000)	Pop. (2010)	Pct. 65+ (2000, %)	Δ Share 65+ (pp)
1	Lee County, Florida	440,888	618,754	25.43	-1.98
2	Indian River County, Florida	112,947	138,028	29.19	-2.02
3	Benton County, Arkansas	153,406	221,339	14.32	-2.13
4	Richmond city, Virginia	197,790	204,214	13.21	-2.13
5	Lake County, Florida	210,528	297,052	26.41	-2.23
6	Pinal County, Arizona	179,727	375,770	16.23	-2.37
7	St. Louis city, Missouri	348,189	319,294	13.74	-2.72
8	St. Lucie County, Florida	192,695	277,789	22.71	-2.77
9	Hernando County, Florida	130,802	172,778	30.85	-5.08
10	Pasco County, Florida	344,765	464,697	26.80	-6.09

Note: Limited to counties with population $\geq 100,000$ in 2000.

Table 6: Bottom 10 Counties by Change in Elderly Population Count (2000-2010)

	County	Pop. (2000)	Pop. (2010)	Pop. 65+ (2000)	Δ Count 65+
1	Philadelphia County, Pennsylvania	1,517,550	1,526,006	213,722	-28,413
2	Allegheny County, Pennsylvania	1,281,666	1,223,348	228,416	-23,357
3	Orleans Parish, Louisiana	484,674	343,829	56,653	-19,014
4	Cuyahoga County, Ohio	1,393,978	1,280,122	217,161	-18,620
5	Wayne County, Michigan	2,061,162	1,820,584	248,982	-18,279
6	Pinellas County, Florida	921,482	916,542	207,563	-13,464
7	Baltimore city, Maryland	651,154	620,961	85,921	-13,109
8	St. Louis city, Missouri	348,189	319,294	47,842	-12,667
9	Milwaukee County, Wisconsin	940,164	947,735	121,684	-12,551
10	Broward County, Florida	1,623,018	1,748,066	261,109	-11,685

Note: Limited to counties with population $\geq 100,000$ in 2000.

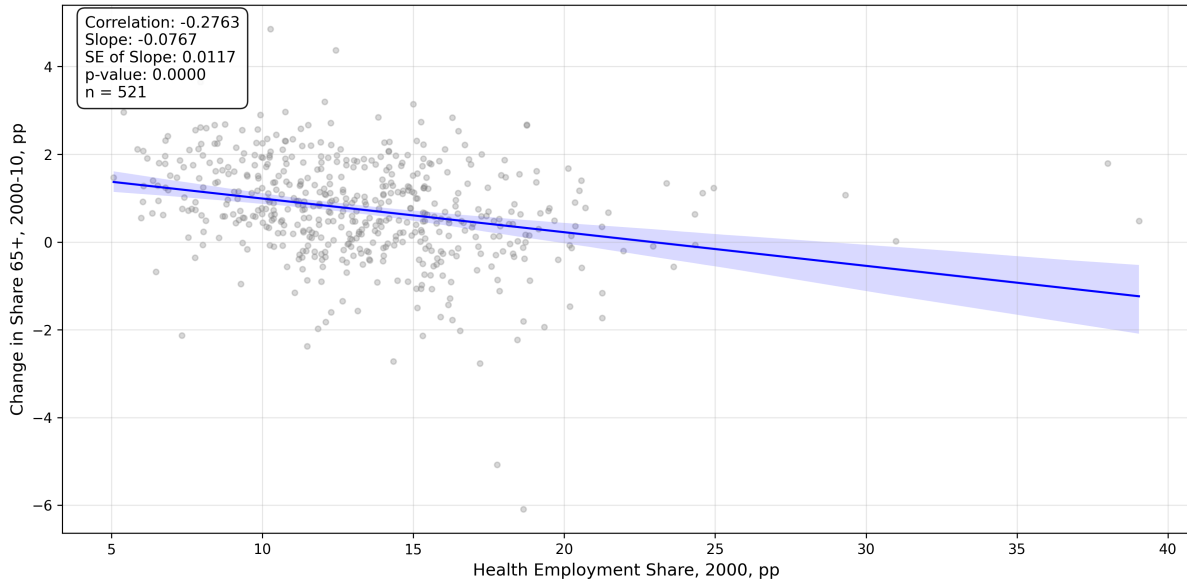


Figure 4: Correlation of healthcare employment share (2000) and change in elderly population share, (2000-2010), restricted to counties with 2000 population of 100K or more. Counties are observations.

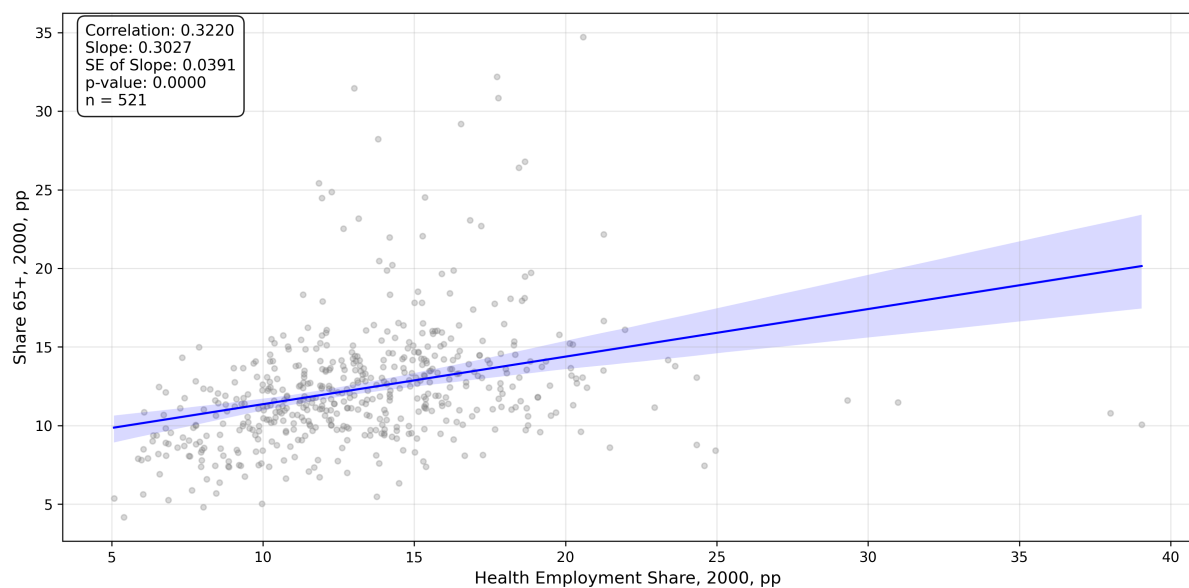


Figure 5: Correlation of healthcare employment share and elderly population share, 2000, restricted to counties with 2000 population of 100K or more. Counties are observations.

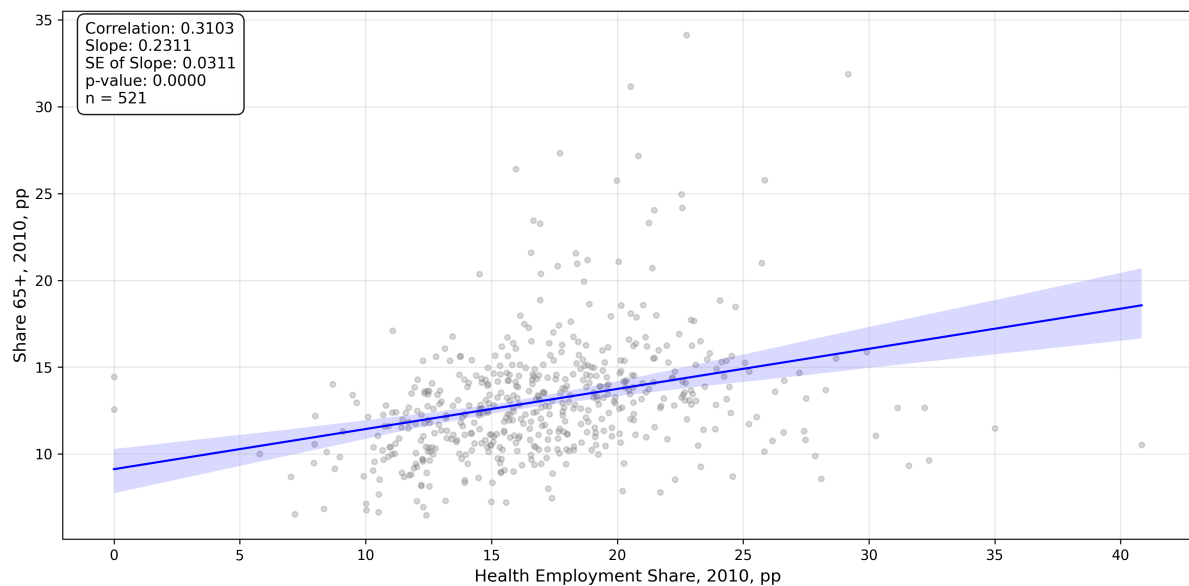


Figure 6: Correlation of healthcare employment share and elderly population share, 2010, restricted to counties with 2000 population of 100K or more. Counties are observations.

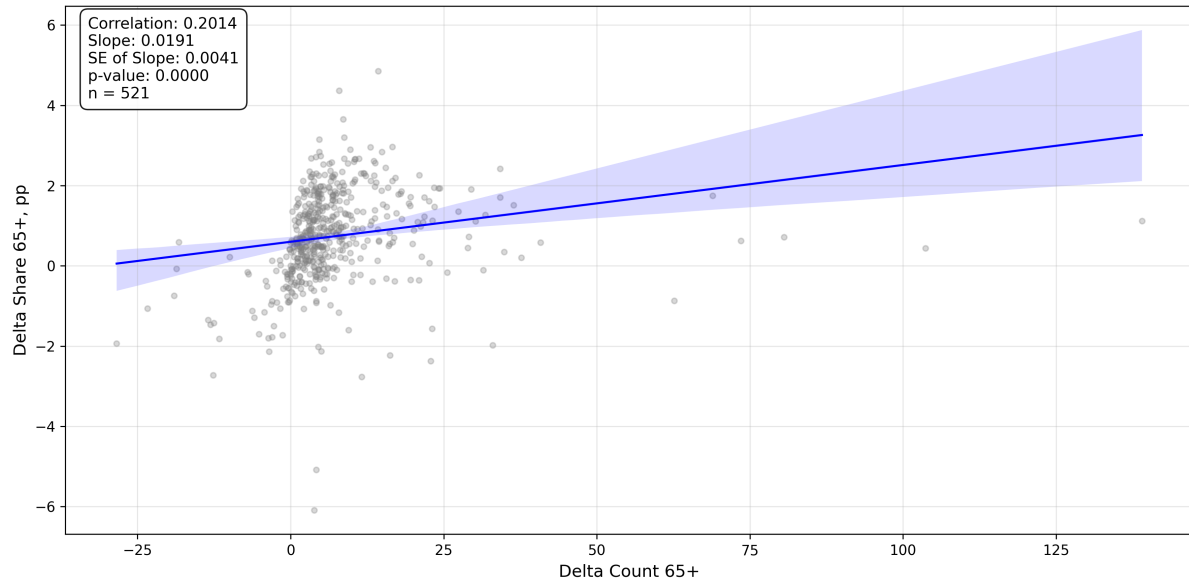


Figure 7: Correlation of changes in counts and shares of the 65+ population, restricted to counties with 2000 population of 100K or more. Counties are observations.

B Appendix on Rural Hospital Closures

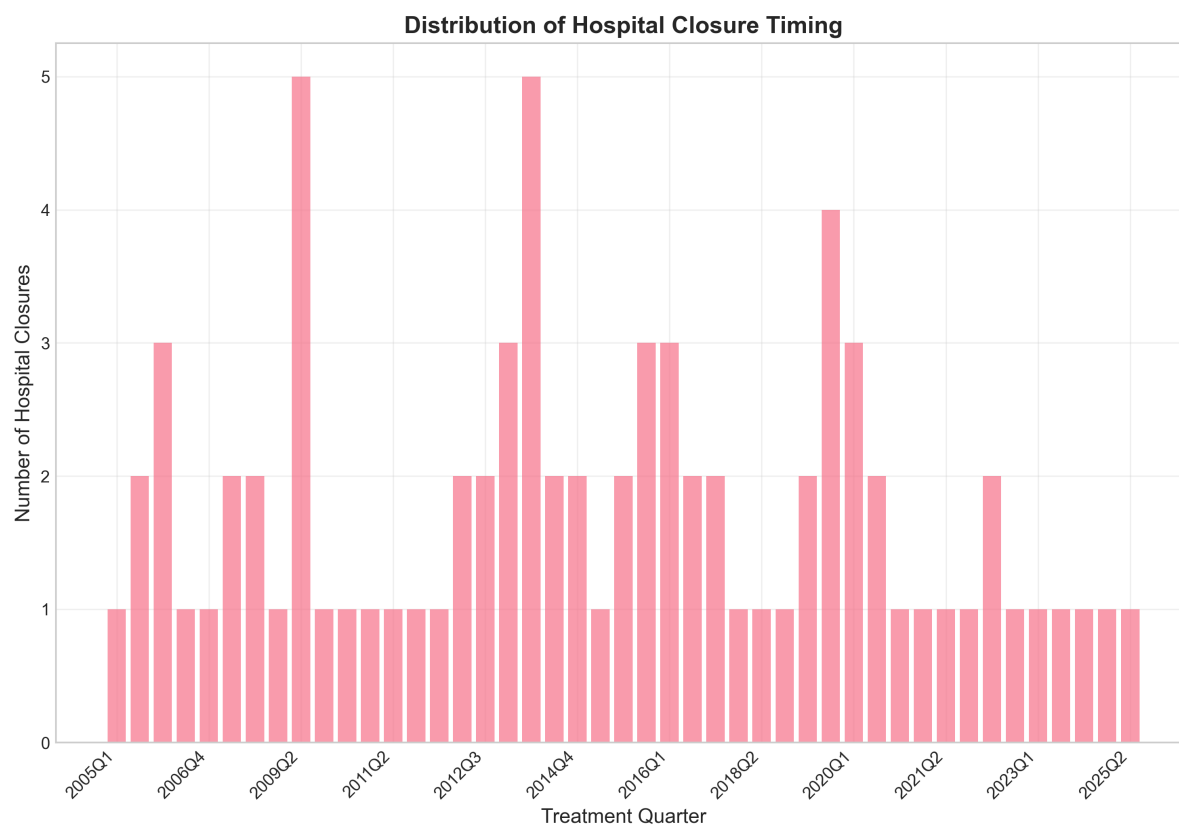


Figure 8: Dates of rural hospital closures, 2005-2025.

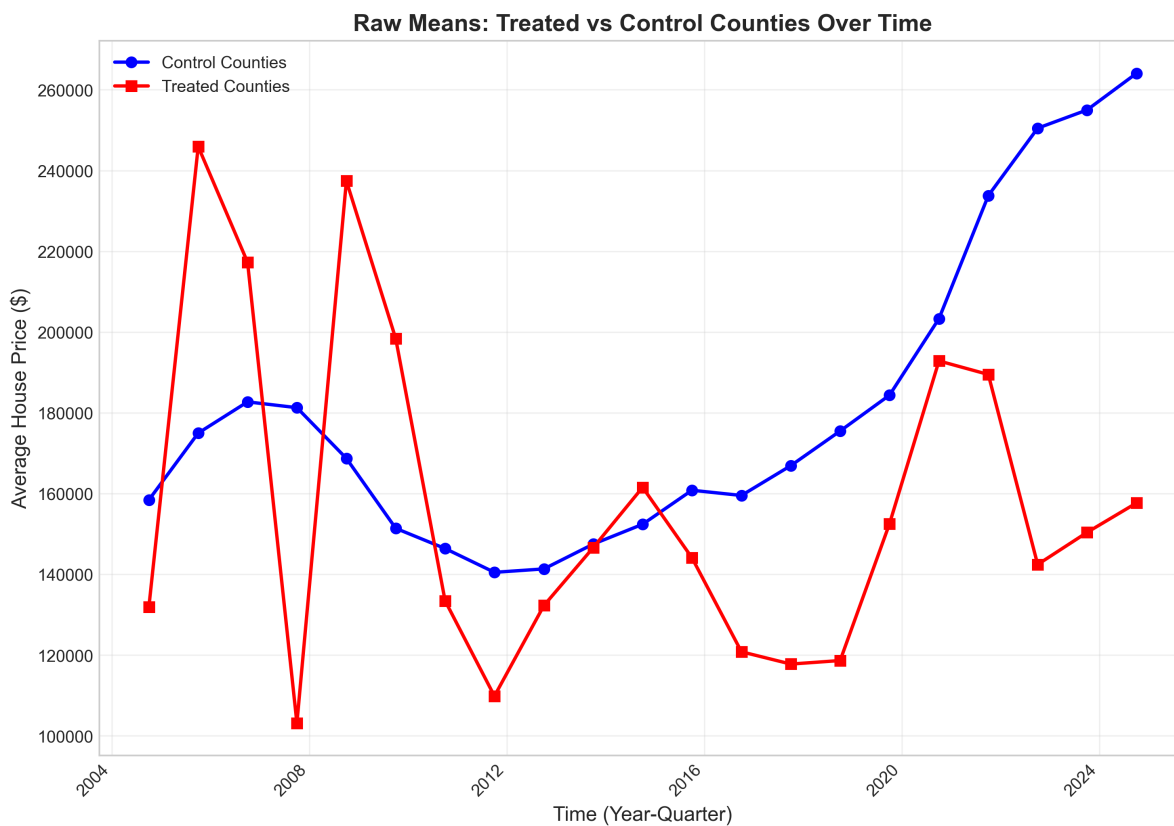


Figure 9: Simple mean across counties of Zillow Home Value Index, by quarter, for treated and control counties. The treated counties are those that experienced exactly one hospital closure during the sample period. The control counties that never experienced any hospital closure during the sample period.

C Literature

(Badilla Maroto et al., 2024) seems closest to the topics I’m interested in. They find a positive impact on housing prices. One of their main findings is shown in Fig. 10.

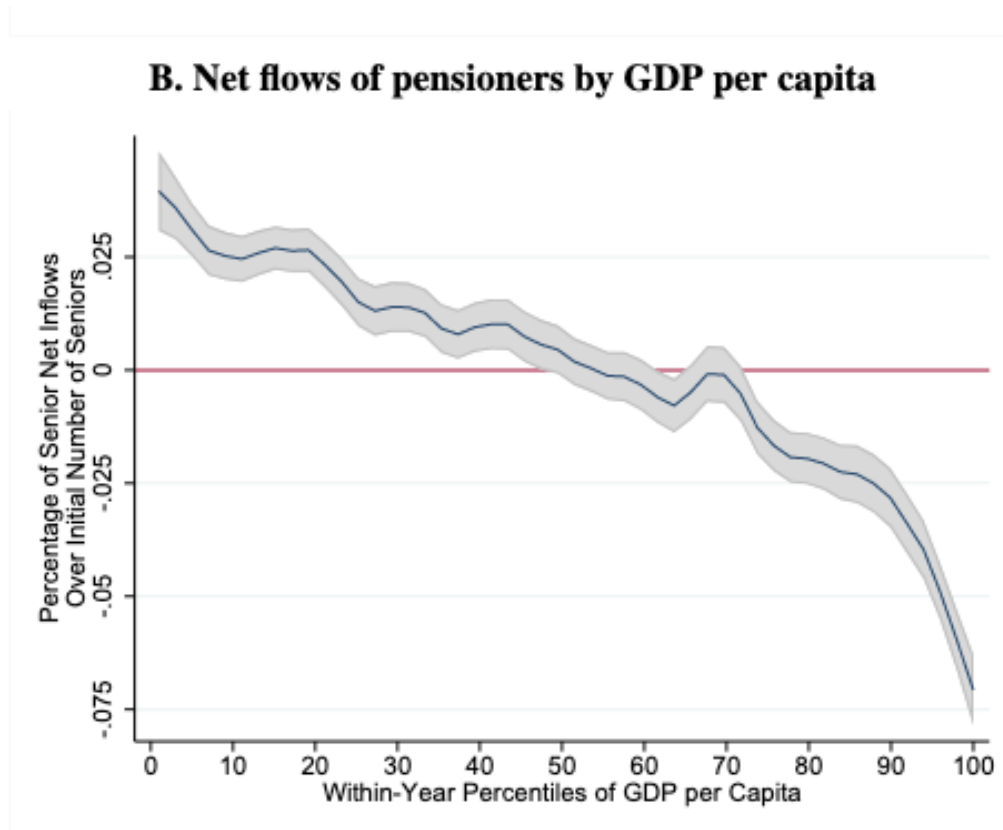


Figure 10: Negative correlation between GDP per capita and inflows of seniors

They use French data because it has microdata on migration flows since 1962, containing “individual-level information on precise residence location, age, marital status, occupation and housing status, and mobility since the last Census wave.”

What I hope to do differently is show specific mechanisms determining senior migration choice, such as presence of healthcare or cost-of-living, and perhaps endogenize choices by local governments to make themselves more or less attractive to retirees (this paper is very reduced-form).

References

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